

L7: Divisive Hierarchical Community Detection

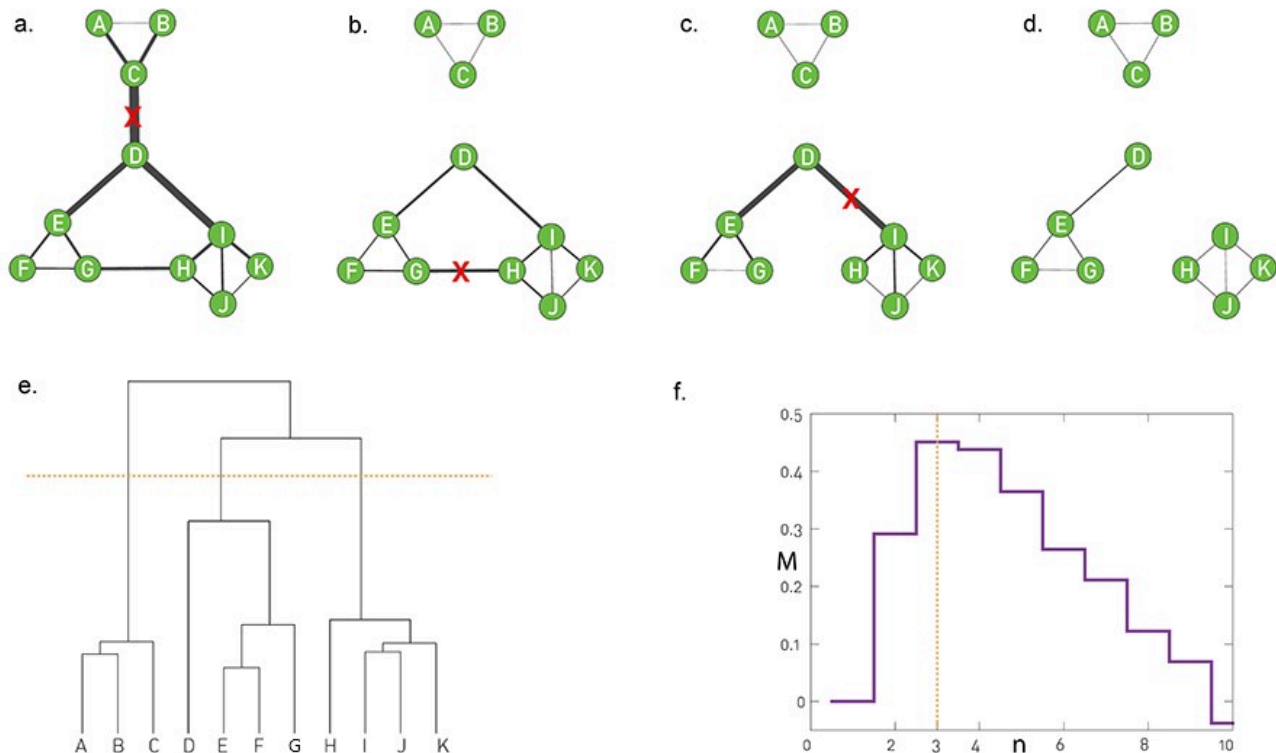


Image 9.12 from networksciencebook.com "The final steps of a divisive algorithm mirror those we used in agglomerative clustering: 1. Compute the centrality x_{ij} of each link. 2. Remove the link with the largest centrality. In case of a tie, choose one link randomly. 3. Recalculate the centrality of each link for the altered network. 4. Repeat steps 2 and 3 until all links are removed.

To illustrate the iterative top-down process followed by divisive hierarchical community detection algorithms, let us focus on the edge betweenness centrality metric.

The Girvan-Newman algorithm uses that metric to remove a single edge in each iteration – the edge with the highest edge betweenness centrality.

The value of the edge betweenness centrality of the remaining edges has to be re-computed because the set of shortest paths changes in each iteration.

The visualizations (a) through (d) show how a toy network changes in four steps of the algorithm, after removing three successive edges. The removal of the first edge, between C and D, creates the first split in the dendrogram, starting at the root. At the left, we have a community formed by A, B, and C – while at the right we have a community with all other nodes.

The next split takes place after we remove two edges, the edge between G and H and the edge between D and I. At that branching point the dendrogram shows three communities: (A,B,C), (D,E,F,G) and (H,I,J,K) (as shown by the horizontal yellow line).

The process can continue, removing one edge at a time, and moving down the dendrogram, until we end up with isolated nodes.

Note that a hierarchical clustering process does *NOT* tell us what is the best set of communities – each horizontal cut of the dendrogram corresponds to a different set of communities. So we clearly need an additional objective or criterion to select the best such set of communities. Such a metric, called modularity M , is shown in the visualization (f), which suggests we cut the dendrogram at the point of three communities (yellow line). We will discuss the metric M a bit later in this lesson.

This algorithm always detect communities in a given network. So, even a random network can be split in this hierarchical manner, even though the resulting communities may not have any statistical significance.

Food For Thought

How would you check if the detected communities have statistical significance, so that a random network does not have any community structure?